

Advanced Kotlin Techniques for Spring Developers

Pasha Finkelshteyn 

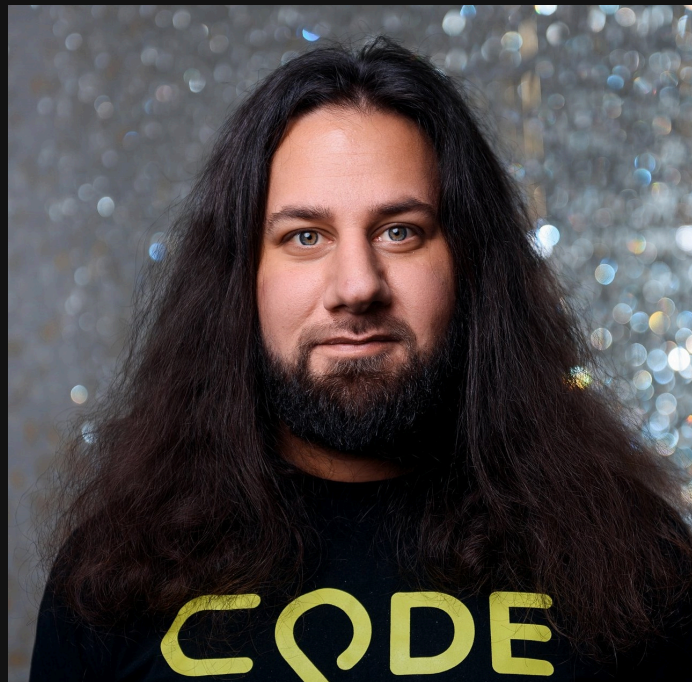
bellsoft

whoami



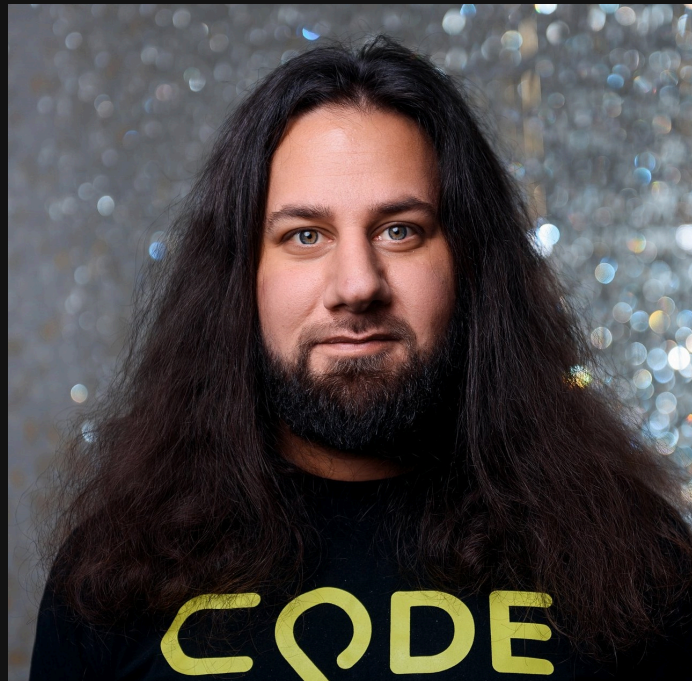
whoami

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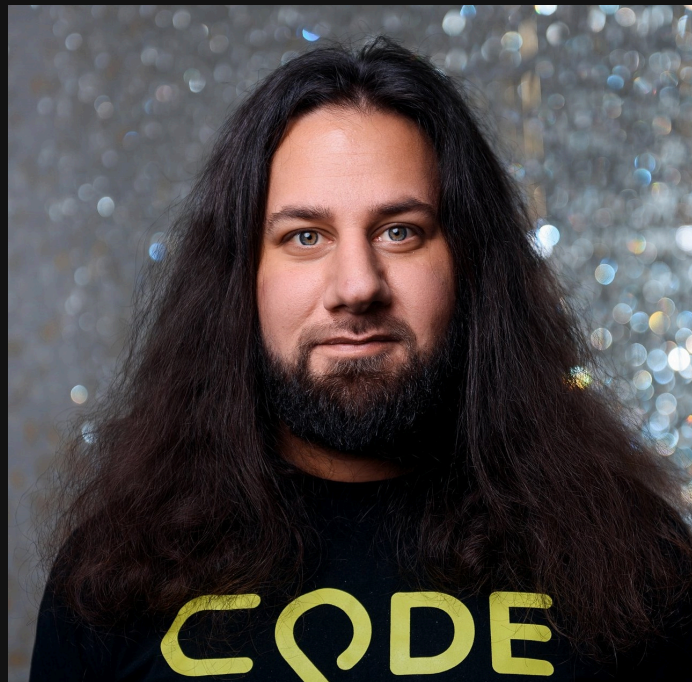
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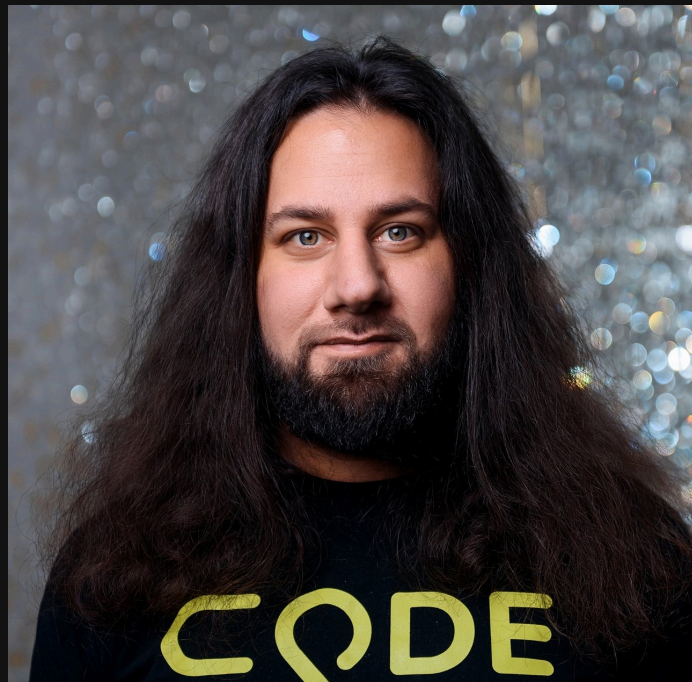
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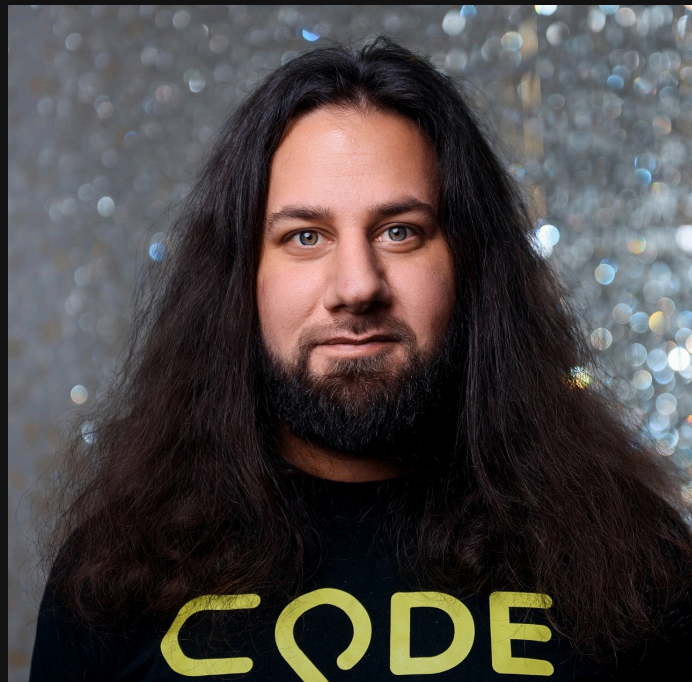
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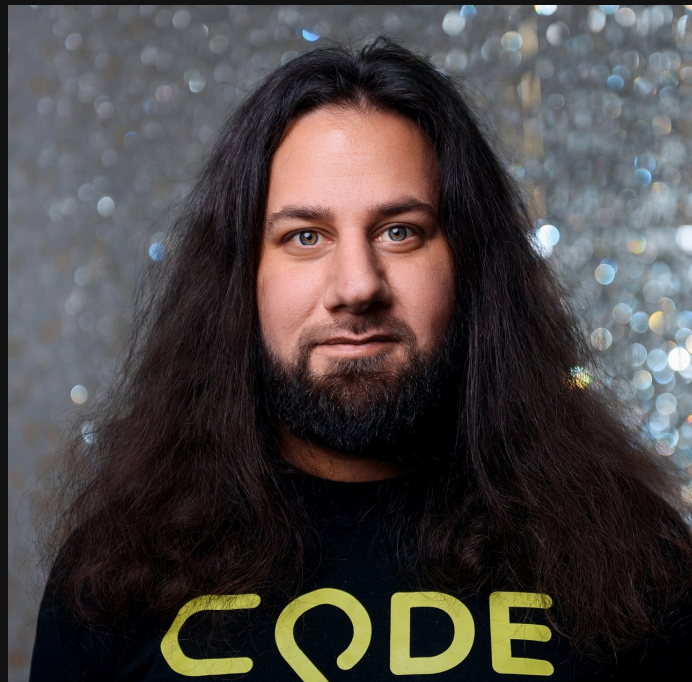
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- 📧 @asm0dey@fosstodon.org



BellSoft

- Vendor of Liberica JDK
- Contributor to the OpenJDK
- Author of ARM32 support in JDK

Liberica is the JDK officially recommended by 



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We know our stuff!



**I hope to showcase
something you don't
know yet!**

My application

My application

- Simple nano-service

My application

- Simple nano-service
- MVC

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- Simple nano-service
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- Validation

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My application

- Simple nano-service
- MVC
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- JPA
- JDBC
- Tests

Where do I start?

<https://start.spring.io>

Project

- ☐ Gradle - Groovy ☒ Gradle - Kotlin
- ☐ Maven

Language

- ☐ Java ☒ Kotlin ☐ Groovy

Spring Boot

- ☐ 3.1.1 (SNAPSHOT) ☒ 3.1.0 ☐ 3.0.8 (SNAPSHOT) ☐ 3.0.7
- ☐ 2.7.13 (SNAPSHOT) ☐ 2.7.12

Minimum dependencies

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Data JPA

SQL

Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.

Validation

I/O

Bean Validation with Hibernate validator.

Spring Web

WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Security

SECURITY

Highly customizable authentication and access-control framework for Spring applications.

PostgreSQL Driver

SQL

A JDBC and R2DBC driver that allows Java programs to connect to a PostgreSQL database using standard, database independent Java code.

Testcontainers

TESTING

Provide lightweight, throwaway instances of common databases, Selenium web browsers, or anything else that can run in a Docker container.

2 files are generated

- `build.gradle.kts`
- `SpringKotlinStartApplication.kt`

What happens

```
import org.jetbrains.kotlin.gradle.tasks.KotlinCompile

plugins {
    id("org.springframework.boot") version "3.3.0"
    id("io.spring.dependency-management") version "1.1.5"
    kotlin("jvm") version "1.9.24"
    kotlin("plugin.spring") version "1.9.24"
    kotlin("plugin.jpa") version "1.9.24"
}

group = "com.github.as00dey"
version = "0.0.1-SNAPSHOT"

java {
    sourceCompatibility = JavaVersion.VERSION_21
}

repositories {
    mavenCentral()
```

The main class

Main class

```
1  import org.springframework.boot.autoconfigure.SpringBootApplication
2  import org.springframework.boot.runApplication
3
4  @SpringBootApplication
5  class SampleApplication
6
7  fun main(args: Array<String>) {
8      runApplication<SampleApplication>(*args)
9  }
```

Main class

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```

runApplication

```
1 inline fun <reified T : Any> runApplication(vararg args: String) =  
2     SpringApplication.run(T::class.java, *args)
```

runApplication

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1 inline fun <reified T : Any> runApplication(vararg args: String) =  
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```

The first goodie of Spring for Kotlin

Let's start implementing

Chapter 1. MVC + Validation

First controller

```
1  @RestController
2  @RequestMapping("/person")
3  class PersonController {
4      @PostMapping
5      fun createPerson(@RequestBody @Valid person: Person) {}
6  }
```

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6  }
```

Person.kt :

```
data class Person(
    val name: String,
    val age: Int
)
```

Make an empty POST ...

```
1  POST localhost:8080/person
2  Content-Type: application/json
```

Make an empty POST ...

```
1 POST localhost:8080/person
2 Content-Type: application/json
```

Make an empty POST ...

```
1 HTTP/1.1 400 Bad Request
2 Content-Type: application/json
3 {
4   "timestamp": 1674735741056,
5   "status": 400,
6   "error": "Bad Request",
7   "path": "/person"
8 }
```

Make an empty POST ...

```
1 HTTP/1.1 400 Bad Request
2 Content-Type: application/json
3 {
4   "timestamp": 1674735741056,
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Make an empty POST ...

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1 HTTP/1.1 400 Bad Request
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3 {
4   "timestamp": 1674735741056,
5   "status": 400,
6   "error": "Bad Request",
7   "path": "/person"
8 }
```

Since `Person` is non-nullable — it's validated without `@NotNull` annotation

Why? How?

build.gradle.kts :

```
1  tasks.withType<KotlinCompile> {  
2      kotlinOptions {  
3          freeCompilerArgs = listOf("-Xjsr305=strict")  
4          jvmTarget = "21"  
5      }  
6  }
```

Why? How?

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```

JSR 305: Annotations for Software Defect Detection:

Nullness annotations (e.g., @NotNull and @CheckForNull)

Internationalization annotations, such as @NonNls or @Nls

Non-empty POST with empty properties

```
1  POST localhost:8080/person
2  Content-Type: application/json
3
4  {"name": null, "age": null}
```

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On client

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1  HTTP/1.1 400 Bad Request
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On client

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1  HTTP/1.1 400 Bad Request
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On server

Non-empty POST with empty properties

```
1 POST localhost:8080/person
2 Content-Type: application/json
3
4 {"name": null, "age": null}
```

On client

```
1 HTTP/1.1 400 Bad Request
2 Content-Type: application/json
```

On server

```
1 ...Instantiation of [simple type, class com.github.asm0dey.sample.Person]
2   value failed for JSON property name due to missing
```

Non-empty POST with empty properties

```
1 POST localhost:8080/person
2 Content-Type: application/json
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4 {"name": null, "age": null}
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On client

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1 HTTP/1.1 400 Bad Request
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On server

```
1 ...Instantiation of [simple type, class com.github.asm0dey.sample.Person]
2   value failed for JSON property name due to missing
```

POST with non-empty name

```
1  POST localhost:8080/person
2  Content-Type: application/json
3
4  {"name": "Pasha", "age": null}
```

POST with non-empty name

```
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2  Content-Type: application/json
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POST with non-empty name

```
1 POST localhost:8080/person
2 Content-Type: application/json
3
4 {"name": "Pasha", "age": null}
```

```
1 HTTP/1.1 200
2 Content-Length: 0
```

POST with non-empty name

```
1 POST localhost:8080/person
2 Content-Type: application/json
3
4 {"name": "Pasha", "age": null}
```

```
1 HTTP/1.1 200
2 Content-Length: 0
```

POST with non-empty name

```
1 POST localhost:8080/person
2 Content-Type: application/json
3
4 {"name": "Pasha", "age": null}
```

```
1 HTTP/1.1 200
2 Content-Length: 0
```



Rechecking

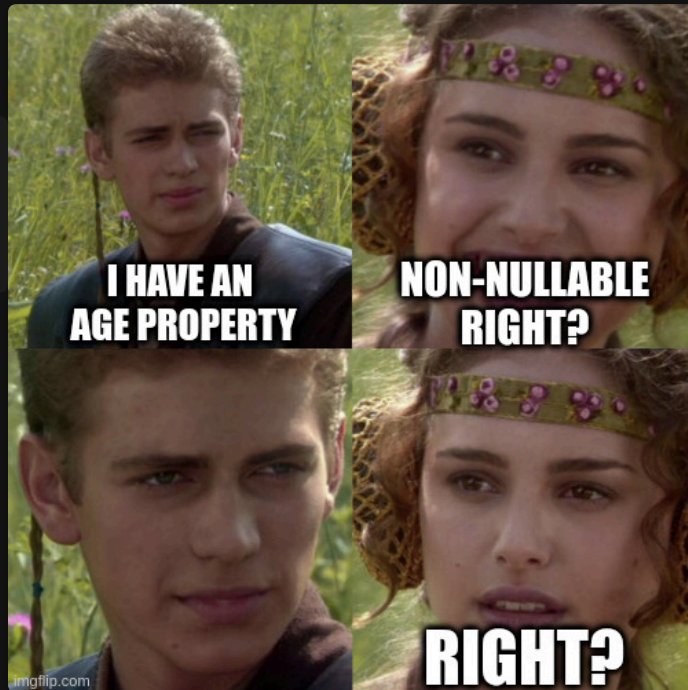
```
1  data class Person(  
2      val name: String,  
3      val age: Double  
4  )
```

Rechecking

```
1  data class Person(  
2      val name: String,  
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```

Rechecking

```
1 data class Person(  
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4 )
```



In JVM primitive types have default values

These types will be JVM primitives:

- Double
- Int
- Float
- Char
- Short
- Byte
- Boolean

Updating class

```
1  data class Person(  
2      val name: String,  
3      @field:NotNull val age: Double?  
4  )
```

Updating class

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4 {"name": "Pasha", "age": null}
```

```
1 HTTP/1.1 400 Bad Request  
2 ...  
3 { "timestamp": 1674760360096, "status": 400, "error": "Bad Request", "path": "/person" }
```

Updating class

```
1 data class Person(  
2     val name: String,  
3     @field:NotNull val age: Double?  
4 )
```

```
1 POST localhost:8080/person  
2 Content-Type: application/json  
3  
4 {"name": "Pasha", "age": null}
```

```
1 HTTP/1.1 400 Bad Request  
2 ...  
3 { "timestamp": 1674760360096, "status": 400, "error": "Bad Request", "path": "/person" }
```

```
1 Field error in object 'person' on field 'age': rejected value [null]
```

Also...

```
1  spring:
2    jackson:
3      deserialization:
4        FAIL_ON_NULL_FOR_PRIMITIVES: true
```

Quick summary

- `-Xjsr305=strict` will make the validation easier
- For JVM primitive types we have to put `@field:NotNull` and mark them nullable
- Sometimes can work it around with jackson settings

JPA

Chapter 2

Nanoentity

```
1  @Entity
2  data class Person(
3      @Id
4      @GeneratedValue(strategy = IDENTITY)
5      var id: Int? = null,
6      @Column(nullable = false)
7      val name: String,
8      @Column(nullable = false)
9      val age: Int,
10 )
```

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- data class

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- `data class`
- `val name` and `val age`

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- `data class`
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Improving

data classes have `copy`, `equals`, `hashCode`, `copy`, and `componentX` defined

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JPA won't be able to write to `val`

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```

JPA won't be able to write to `val`

But there is no no-arg constructor!

How to make it work?

Magic:

```
1  kotlin("plugin.jpa") version "1.8.0"
```

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How to make it work?

Magic:

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1  kotlin("plugin.jpa") version "1.8.0"
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- Puts annotations on the fields
- Adds a default constructor in bytecode*

But there is no no-arg constructor!

How to make it work?

Magic:

```
1  kotlin("plugin.jpa") version "1.8.0"
```

- Puts annotations on the fields
- Adds a default constructor in bytecode*

* In Kotlin the default constructor would not be possible, but in Java it is

Current result

```
1  @Entity
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5      var id: Int? = null,
6      @Column(nullable = false)
7      var name: String,
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9      var age: Int,
10 )
```

Is this enough?

Not quite.

At the very least we have to redefine `equals` and `hashCode` .

For example...

```
1  @Entity
2  class Person(
3      // properties
4  ) {
5      // equals...
6      override fun hashCode(): Int {
7          return id ?: 0
8      }
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```

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JDBC

Chapter 3

Obtain user by id

Let's imagine we need to call the following:

```
1  SELECT *  
2  FROM users  
3  WHERE id = ?
```

Or, for example...

Or, for example...

In Java

☕ Let's inline mapper

```
public List<Person> findById(int id) {  
    return jdbcTemplate.query("SELECT * FROM users WHERE id = ?", new UserRowMapper(), id);  
}  
  
private static class UserRowMapper implements RowMapper<Person> {  
    @Override  
    public Person mapRow(ResultSet resultSet, int i) throws SQLException {  
        int id = resultSet.getInt("id");  
        String name = resultSet.getString("name");  
        Double age = resultSet.getDouble("age");  
        return new Person(id, name, age);  
    }  
}
```

In Java



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        String name = resultSet.getString("name");  
        Double age = resultSet.getDouble("age");  
        return new Person(id, name, age);  
    }  
}
```

In Java

☕ Let's inline mapper

```
public List<Person> findById(int id) {  
    return jdbcTemplate.query("SELECT * FROM users WHERE id = ?", new UserRowMapper(), id);  
}  
  
private static class UserRowMapper implements RowMapper<Person> {  
    @Override  
    public Person mapRow(ResultSet resultSet, int i) throws SQLException {  
        int id = resultSet.getInt("id");  
        String name = resultSet.getString("name");  
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        return new Person(id, name, age);  
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In Java

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public List<Person> findById(int userId) {  
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In Java



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In Java



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        return new Person(id1, name, age);  
    }, userId);  
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```

I don't like it

- Too many mappers
- Parameters are too far from query



Why should it be so?

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Let's Look at the signature

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```
1 public <T> List<T> query(String sql, RowMapper<T> rowMapper, @Nullable Object... args)
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```
1 public <T> List<T> query(String sql, RowMapper<T> rowMapper, @Nullable Object... args)
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Because in  vararg can be only the last... 

JdbcTemplate in Kotlin

```
1  return jdbcTemplate.query("SELECT * FROM users WHERE id = ?", userId) { rs, _ ->
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6  }
```

- `vararg` doesn't have to be in the last position
- unused parameter of a lambda can be named `_`

Extension functions

```
1 fun <T> JdbcOperations.query(  
2     sql: String,  
3     vararg args: Any,  
4     function: (ResultSet, Int) -> T  
5 ): List<T>
```

Extension functions

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Which allows

```
1 return jdbcTemplate.query("SELECT * FROM users WHERE id = ?", userId)  
2 { rs, _ ->  
3     // TODO: ResultSet -> Person  
4 }
```

You can do the same for your own code!

From 

```
1 public List<String> transformStrings(Function<String, String> mapper, String... args){}
```

You can do the same for your own code!

From 

```
1 public List<String> transformStrings(Function<String, String> mapper, String... args){}
```

To 

```
1 fun transformStrings(vararg args: String, mapper: (String) -> String): List<String>{}
```

More on extensions for Spring

- [spring-beans](#)
- [spring-context](#)
- [spring-core](#)
- [spring-jdbc](#)
- [spring-messaging](#)
- [spring-r2dbc](#)
- [spring-test](#)
- [spring-tx](#)
- [spring-web](#)
- [spring-webflux](#)
- [spring-webmvc](#)

Reactive persistence

Chapter 4

Who knows what is reactive?



Who knows what is reactive?

- non-blocking



Who knows what is reactive?

- non-blocking
- asynchronous



Who knows what is reactive?

- non-blocking
- asynchronous
- handles back-pressure



Who knows what is reactive?

- non-blocking
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The result is a monad...



Who knows what is reactive?

- non-blocking
- asynchronous
- handles back-pressure

The result is either:

- **Mono** : produces 0 to 1 items when it's ready
- **Flux** : Produces 0 to ∞ items



In Kotlin

Coroutines operate on `suspend` functions

`suspend` can be called from:

- coroutine context
- another `suspend` `fun`

returns usual types - Lists, Strings, etc

```
1  suspend fun randomNum(): Int {  
2      val x = service1.randomNum() // suspend  
3      val y = service2.randomNum() // suspend  
4      return x + y  
5  }
```

In Kotlin

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Spring R2DBC

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@Repository
class Repo(connectionFactory: ConnectionFactory) {
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class Repo(connectionFactory: ConnectionFactory) {
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        client
            .sql("INSERT INTO users (name, email, age) VALUES ('Pasha', :email, NULL) RETURNING id")
            .bind("email", RandomStringUtils.randomAlphabetic(20))
            .fetch()
            .awaitSingle()["id"] as? Long ?: error("not long on not returned")
}
```

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            .bind("email", RandomStringUtils.randomAlphabetic(20))
            .fetch()
            .awaitSingle()["id"] as? Long ?: error("not long on not returned")
    // inside Spring ↓
    suspend fun <T> RowsFetchSpec<T>.awaitSingle(): T {
        return first().awaitSingleOrNull() ?: throw EmptyResultDataAccessException(1)
    }
}
```

Spring R2DBC

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```

Transactions

```
1  @Transactional
2  suspend fun failTransactional(): Long {
3      val curId = repo.createUserAndReturnId()
4      repo.createUserWithConflictingId(curId) // will throw
5      return curId
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```

Nothing changes from the client standpoint!

Repositories

```
interface User : CoroutineCrudRepository<User, Long> {  
}
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Repositories

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interface User : CoroutineCrudRepository<User, Long> {  
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// Inside Spring  
interface CoroutineCrudRepository<T, ID> : Repository<T, ID> {  
    // ...  
    suspend fun <S : T> save(entity: S): T  
    // Flow is an async unbounded collection  
    fun findAll(): Flow<T>  
    // ...  
}
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Repositories

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Repositories

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interface User : CoroutineCrudRepository<User, Long> {  
    suspend fun findOne(id: String): User  
    fun findByFirstname(firstname: String): Flow<User>  
    suspend fun findAllByFirstname(id: String): List<User>  
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Function is either suspend or returns `Flow`

Transactions with repositories?

```
1  @Transactional
2  suspend fun failTransactional(): Long {
3      val u = User(null, "me@asm0dey.site", 37)
4      val savedId = repo.save(u)
5      repo.save(u.copy(id = savedId)) // will throw
6      return curId
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```

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Configuration

Let's start simple

```
1  val beans = beans {  
2      bean { jacksonObjectMapper() }  
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Modified Jackson's `ObjectMapper` to work with `data` classes from `jackson-module-kotlin`

```
1  @Bean  
2  fun kotlinMapper(): ObjectMapper {  
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```

4 lines instead of 1 🤖

Custom bean

```
1  class JsonLogger(private val objectMapper: ObjectMapper) {  
2      fun log(o: Any) {  
3          if (o::class.isData) {  
4              println(objectMapper.writeValueAsString(o))  
5          } else println(o.toString())  
6      }  
7  }  
8  
9  val beans = beans {  
10     bean { jacksonObjectMapper() }  
11     bean(::JsonLogger)  
12 }
```

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Arbitrary logic

```
1  val beans = beans {  
2    bean { jacksonObjectMapper() }  
3    bean(::JsonLogger)  
4    bean("randomGoodThing", isLazyInit = Random.nextBoolean()) {  
5      if (Random.nextBoolean()) "Norway" else "Well"  
6    }  
7  }
```

Arbitrary logic

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OK How do I use it?

Let's return to our very first file

```
1  runApplication<SampleApplication>(*args)
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Let's change it to

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```

And run it...

```
1  Started SampleApplicationKt in 1.776 seconds (process running for 2.133)
```

Let's test it

Bean:

```
1  @Component
2  class MyBean(val jsonLogger: JsonLogger) {
3      fun test() = jsonLogger.log("Test")
4  }
```

Test:

```
1  @SpringBootTest
2  class ConfigTest {
3      @Autowired private lateinit var myBean: MyBean
4      @Test
5      fun testIt() = assertEquals("Test", myBean.test())
6  }
```

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Let's test it

Bean:

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1  @Component
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That's because our tests do not call
`main` !



Requires some glue to work

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Security

Spring Security

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Thank you!

Thank you! Questions?

-  asm0di0
-  @asm0dey@fosstodon.org
-  me@asm0dey.site
-  asm0dey
-  asm0dey
-  asm0dey
-  asm0dey
-  asm0dey



END